Performance Analysis of IPsec IKEv2 Rekeying Interval in 5G Backhaul Environments*

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Abstract

In 5G mobile networks, IPsec with IKEv2 is mandated to secure the backhaul segment per 3GPP TS 33.501. However, periodic CHILD_SA rekeying introduces cryptographic overhead that may degrade performance, and 3GPP does not specify a recommended interval. In this paper, we analyze the impact of varying rekeying intervals on throughput and TCP retransmissions in a 5G backhaul environment, and derive guidelines for selecting an optimal interval.

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